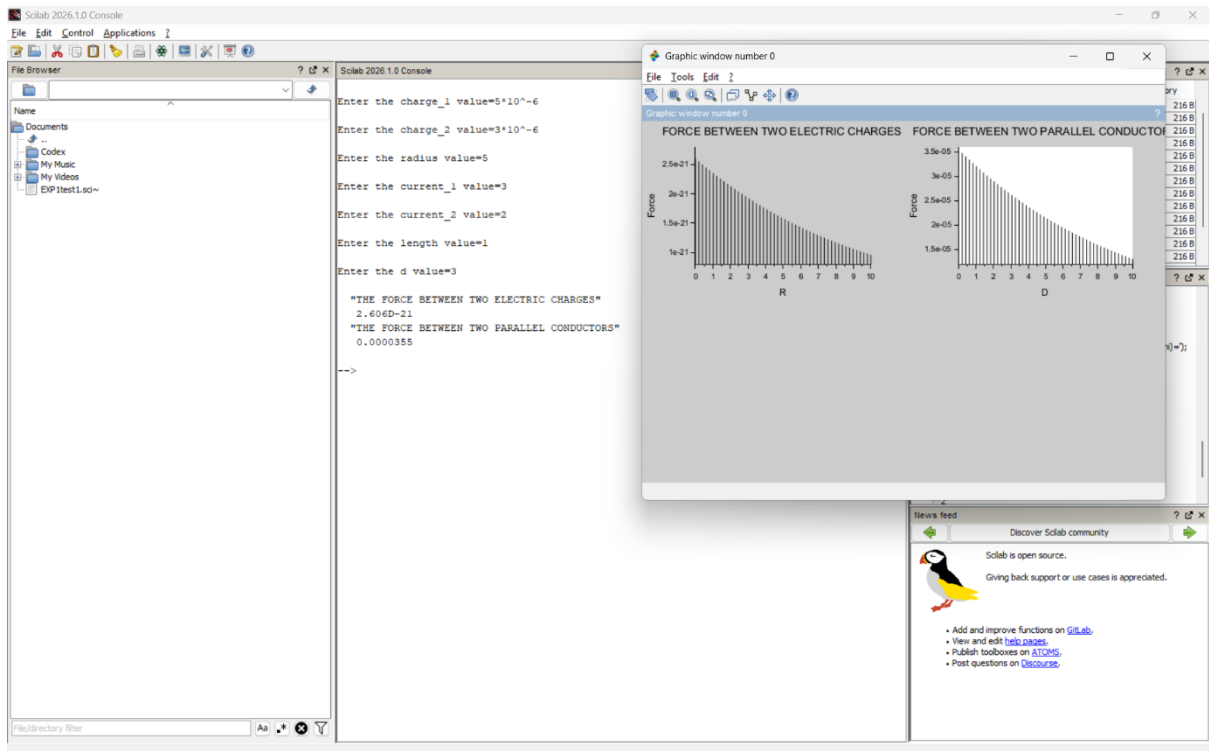



```

1 clear;
2 clear;
3 charge_1 = input("Enter the charge_1 value=");
4 charge_2 = input("Enter the charge_2 value=");
5 radius = input("Enter the radius value=");
6 current_1 = input("Enter the current_1 value=");
7 current_2 = input("Enter the current_2 value=");
8 l = input("Enter the length value=");
9 d = input("Enter the d value=");
10 mu = 4*pi*10^-7;
11 epsilon=0.85*10^-12;
12 pi=3.14;
13 F=(charge_1 * charge_2)/(pi*epsilon*radius^2);
14 f=(mu*current_1*current_2*l)/(pi*d);
15 disp("THE FORCE BETWEEN TWO ELECTRIC CHARGES", [F]);
16 disp("THE FORCE BETWEEN TWO PARALLEL CONDUCTORS", [f]);
17 x = linspace(0, 10, 50);
18 y = F * exp(-0.1 * x);
19 z = f * exp(-0.1 * x);
20 figure();
21 subplot(2, 2, 1);
22 plot2d3(x,y);
23 label("F");
24 label("Force");
25 title("FORCE BETWEEN TWO ELECTRIC CHARGES");
26 subplot(2, 2, 2);
27 plot2d3(x,z);
28 label("f");
29 label("Force");
30 title("FORCE BETWEEN TWO PARALLEL CONDUCTORS");

```

Line 30, Column 47



Exp 3:-

charge 1 :- 5×10^{-6}

charge 2 :- 3×10^{-6}

radius :- 5 cm

current 1 :- 3

current 2 :- 2

length :- 1 mm

radius :- 2 cm

B/w two

electric charges

$$F = \left(\frac{q_1 q_2}{4} \right) \times \pi \times \epsilon \times r^2$$

$$= \frac{(5 \times 10^{-6})(3 \times 10^{-6})}{4} \times 3.14 \times 8.854 \times 10^{-12} \times 5^2$$

$$= \frac{15 \times 10^{-12}}{4} \times 3.14 \times 8.854 \times 10^{-12} \times 25$$

$$= 1.0425 \times 10^{-22} \times 25$$

$$\therefore F = 2.606 \times 10^{-21} \text{ N}$$

B/w two parallel conductors

$$f = \left(\frac{\mu I_1 I_2 l}{2} \right) \times \pi \times d$$

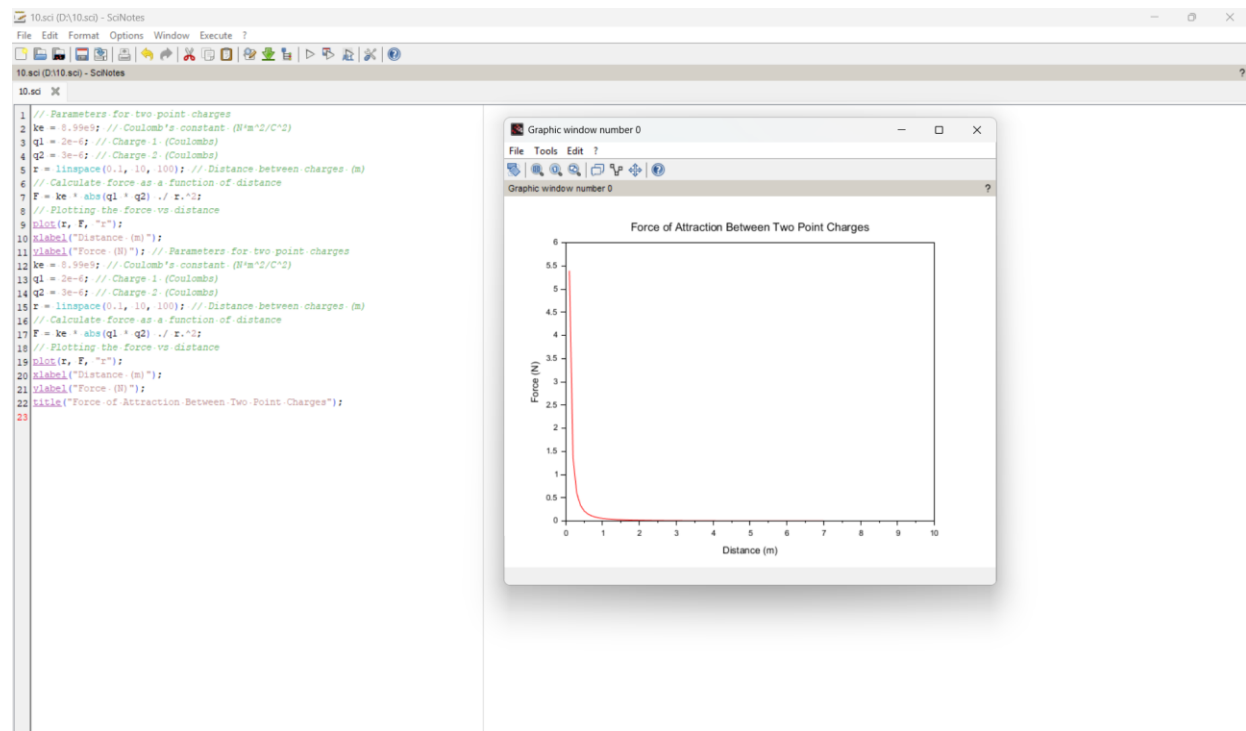
$$f = \frac{(1.256 \times 10^{-6})(3)(2)(1)}{2} \times 3.14 \times 3$$

$$= \frac{7.536 \times 10^{-6}}{2} \times 3.14 \times 3 \Rightarrow 3.768 \times 10^{-6} \times 3.14 \times 3$$

$$= 1.13152 \times 10^{-5} \times 3$$

$$\therefore f = 3.55 \times 10^{-5} \text{ N}$$

Case 1 :Given two point charges $q_1=2\times 10^{-6}$ and $q_2=3\times 10^{-6}$ C, calculate and plot the force of attraction between them as a function of distance, ranging from 0.1 m to 10 m. Explain how the force changes with the distance between the charges.



Case 2:Given two point charges $q_1=5\times 10^{-6}$ C and $q_2=2\times 10^{-6}$, calculate and plot the force of attraction between them as the distance varies from 0.5 m to 5 m. Discuss how the force behaves with the varying distance and charge magnitudes.

